104/Math(C) 23-24/12114

B.Sc. 1st Semester (Honours) Examination, 2023-24

MATHEMATICS

Course ID : 12114 **Course Code : SH-MTH-103/GE-1**

Course Title : Calculus, Geometry & Vector analysis

[New Syllabus-2022]

Time : 2 Hours Full Marks : 25

The figures in the right margin indicate marks.

Candidates are required to answer in their own words as far as practicable.

UNIT-I

1. Answer ***any five*** from the following questions: $5\times2=10$

a) Evaluate: $\lim\limits_{x\to0}\left(\frac{\tan x-x}{x-\sin x}\right)$.

b) Write down Leibnitz's theorem on successive derivative.

c) Find $\int_0^{\frac{\pi}{2}}\cos^5 x\,dx$.

d) Find the centre and radius of the sphere $3x^2+3y^2+3z^2+2x-4y-2z-1=0$.

e) If $\vec{a}\times\vec{b}+\vec{b}\times\vec{c}+\vec{c}\times\vec{a}=\vec{0}$, then show that the vectors $\vec{a},\vec{b},\vec{c}$ are coplanar.

[Turn Over]

f) If $\vec{r} = 3t\,\hat{i} + 3t^2\,\hat{j} + 2t^3\,\hat{k}$ then find $\left[\frac{d\vec{r}}{dt}\ \frac{d^2\vec{r}}{dt^2}\ \frac{d^3\vec{r}}{dt^3}\right]$.

g) Find the nature of the conic $\frac{1}{r} = 23 - 5\cos\theta$.

h) Find the transformed equation of the curve $2x^2 + 4xy + 3y^2 - 2x - 4y + 7 = 0$ when the origin is shifted to the point (2, –3).

UNIT-II

2. Answer ***any four*** from the following questions:

4×5=20

a) If $y = e^{m \sin^{-1} x}$, then show that

$$(1 - x^2)y_{n+2} - (2n + 1)x\,y_{n+1} - (n^2 + m^2)y_n = 0.$$

b) Find the asymptotes of the curve

$$x^3 - 2y^3 + xy(2x - y) + y(x - y) + 1 = 0$$

c) If $I_{m,n} = \int \cos^m x \sin nx\, dx$, then prove that

$$I_{m,n} = -\frac{\cos^m x \cos nx}{m+n} + \frac{m}{m+n} I_{m-1,n-1}.$$

d) i) Prove a necessary and sufficient condition that a vector $\vec{r} = \vec{f}(t)$ to have constant direction is $\vec{f} \times \frac{d\vec{f}}{dt} = \vec{0}$.

ii) If $\vec{r} = 3t\hat{i} + 3t^2\hat{j} + 2t^3\hat{k}$, then find $\frac{d\vec{r}}{dt} \times \frac{d^2\vec{r}}{dt^2}$ at $t=1$. 3+2

e) If one of the straight lines $ax^2 + 2hxy + by^2 = 0$ concides with one of the straight lines $a'x^2 + 2h'xy + b'y^2 = 0$ and remaining two straight lines are at right angles, then show that

$$h\left(\frac{1}{b} - \frac{1}{a}\right) = h'\left(\frac{1}{b'} - \frac{1}{a'}\right).$$

f) i) Evaluate $\int_1^2 \left(\vec{r} \times \frac{d^2\vec{r}}{dt^2}\right) dt$ where $\vec{r} = t^2\hat{i} + 2t\hat{j} + 3t^2\hat{k}$.

ii) Show that $[\vec{a}+\vec{b}, \vec{b}+\vec{c}, \vec{c}+\vec{d}] = 2[\vec{a}\vec{b}\vec{c}]$. 3+2

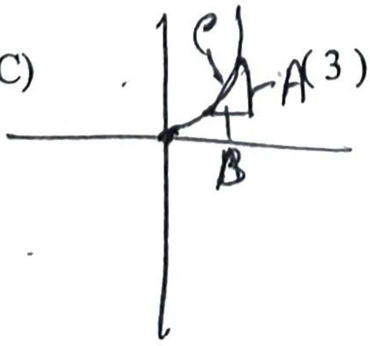
UNIT-III

3. Answer ***any one*** from the following questions:

1×10=10

a) i) Show that the volume of the parallelepiped, whose edges are represented by $3\hat{\imath} + 2\hat{\jmath} - 4\hat{k}$, $3\hat{\imath} + \hat{\jmath} + 3\hat{k}$ and $\hat{\imath} - 2\hat{\jmath} + \hat{k}$, is 49 cubic units.

ii) Find the equation of the cone whose vertex is the point (3,2,1), semi vertical



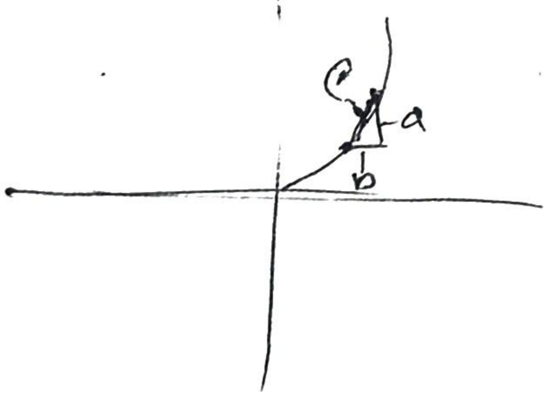
angle 30^0 and axis $\frac{x-3}{1} = \frac{y-2}{4} = \frac{z-1}{3}$.

iii) Prove that origin is a node of the curve $x^3 + y^3 = 32xy$. 4+4+2

b) i) Find the equation of the sphere which passes through the points (0, 0, 0), (0, 1, −1), (−1, 2, 0) and (1,2,3)

ii) Find the envelope of the family of lines $\frac{x}{a}+\frac{y}{b}=1$, where the parameters are connected by $a^2+b^2=c^2$, where c being constant.

iii) Apply the vector method to show that the internal bisector of the angle A of a triangle ABC divides the sides the side BC in the ratio AB:AC . 4+3+3



B.Sc. 1st Semester (Honours) Examination, 2023-24

MATHEMATICS

Course Code : 12114 Course Code : SH-MTH-103/GE-1

Course Title : Calculus, Geometry & Differential Equation

[Old Syllabus-2017]

Time : 2 Hours Full Marks : 40

The figures in the right margin indicate marks.

Candidates are required to answer in their own words as far as practicable.

UNIT-I

1. Answer ***any five*** from the following questions: $5\times2=10$

a) Find the pedal equation of the parabola $y^2=4ax$ w.r.t. its vertex.

b) Find the asymptotes of the hyperbola, $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$.

c) Show that $\frac{1}{3x^3y^3}$ is an integrating factor of $y\left(xy+2x^2y^2\right)dx+x\left(xy-x^2y^2\right)dy=0$.

d) If in the Cauchy's MVT, $f(x)=e^x$ and $F(x)=e^{-x}$, show that c is the Arithmetic Mean between a and b.

e) Obtain the first order ODE having $(x-c)^2+y^2=4$ as its solution.

f) Find the greatest and least values of the function $f(x)=3x^4-2x^3-6x^2+6x+1$ in the interval [0, 2].

g) Evaluate $\lim_{x\to 0}\frac{\cosh x-\cos x}{x\sin x}$.

h) Find the nth derivative of the function $y=\cos^4 x$.

UNIT-II

2. Answer ***any four*** from the following questions: 4×5=20

a) Obtain reduction formula for $\int \sin^n\theta d\theta;\ \int_0^{\pi/2}\sin^n\theta d\theta$.

b) Find the equation of the Cone whose vertex is the point (1,2,3) and guiding curve is the circle $x^2+y^2+z^2=9,\ x+y+z=1$.

c) Find the equation of the right circular cylinder of radius 3 and whose axis is $\frac{x-1}{2}=\frac{y-2}{-3}=\frac{z-3}{6}$.

d) Prove that the straight line $r\cos(\theta-\alpha)=a$ will touch the conic $\frac{l}{r}=1-e\cos\theta$ if $l^2+2ael\cos\alpha+a^2(e^2-1)=0$.

e) Solve $\frac{dy}{dx}+\frac{y}{x}\log y=\frac{y}{x^2}(\log y)^2$.

f) If $y=e^{a\sin^{-1}x}$ then prove that $(1-x^2)y_{n+2}-(2n+1)xy_{n+1}-(n^2-a^2)y_n=0$.

Hence deduce that

$(y_n)_0=\{(n-2)^2+a^2\}\{(n-4)^2+a^2\}......\{(2^2+a^2\}$ if n be even.

UNIT-III

3. Answer ***any one*** from the following questions:

$1\times10=10$

a) i) Reduce the equation $x^2-2xy+y^2+6x-14y+29=0$, to its conical from and hence determine the nature of the curve represented by it. Also find the vertex.

ii) Show that the initial value problem $\left|\frac{dy}{dx}\right|+2|y|=0, y(0)=1$ has no solution.

iii) Solve : $\left(x^2+y^2\right)dx+\left(x^2-xy\right)dy=0$.

5+2+3

b) i) Find the surface of the right circular cylinder of greatest surface which can be inscribed in a sphere of radius r. 7

ii) Find the maxima and minima of the function

$$f(x) = sinx + \frac{1}{2}sin2x + \frac{1}{3}sin3x \quad \forall x \in [0, \pi].$$

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